CHAPTER 7

ALCOHOLS, PHENOLS AND ETHERS

CHAPTER ESSENCE AT A GLANCE:
Synthesis and reactions of alcohols, phenols, and ethers, including acidity comparisons, Kolbe's and Reimer-Tiemann reactions, and Williamson synthesis.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Alcohol Preparation & Hydration:** Alcohols are synthesized industrially by acid-catalyzed hydration of alkenes following Markovnikov addition, reduction of carbonyls, or addition of Grignard reagents to aldehydes and ketones.

• **Comparative Acidity of Phenols:** Phenols are markedly more acidic than alcohols because the resulting phenoxide anion is stabilized by resonance delocalization of negative charge over the aromatic ring, an effect enhanced by electron-withdrawing nitro groups.

• **Electrophilic Functionalization of Phenol:** Phenol exhibits exceptional ortho-para reactivity toward electrophilic substitution, undergoing Kolbe carboxylation to produce salicylic acid and Reimer-Tiemann formylation to yield salicylaldehyde.

• **Lucas Test & Alcohol Classification:** Distinguishes primary, secondary, and tertiary alcohols based on the relative rate of alkyl chloride turbidity formation upon reaction with concentrated hydrochloric acid and anhydrous zinc chloride.

ESSENTIAL FORMULAS & GOVERNING EQUATIONS

$$\mathrm{CH_3CH=CH_2 + H_2O \xrightarrow{H^+} CH_3\text{-}CH(OH)\text{-}CH_3} \quad \text{(Markovnikov)}$$

Acid-catalyzed hydration of alkenes to form secondary alcohols

$$K_a = \frac{[\mathrm{C_6H_5O^-}][\mathrm{H^+}]}{[\mathrm{C_6H_5OH}]} \rightarrow \mathrm{p}K_a(\text{Phenol}) \approx 10 < \mathrm{p}K_a(\text{Ethanol}) \approx 16$$

Acid dissociation equilibrium demonstrating high relative acidity of phenol

$$\mathrm{C_6H_5OH + CHCl_3 + 3NaOH \rightarrow \textit{o}\text{-}HOC_6H_4CHO} \quad \text{(Reimer-Tiemann)}$$

Reimer-Tiemann formylation of phenol yielding salicylaldehyde

$$\mathrm{C_6H_5ONa + CO_2 \xrightarrow{400\ K,\ 4-7\ atm} \textit{o}\text{-}HOC_6H_4COOH} \quad \text{(Kolbe's reaction)}$$

Kolbe's carboxylation of sodium phenoxide to produce salicylic acid

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Cleavage of mixed ethers with HI: If one alkyl group is 3°, iodide attacks 3° carbon (S N1) to form 3° alkyl iodide. If 1° divided by 2°, iodide attacks smaller group (S N2).

• **Alert:** [Valuation Rubric] Step 1 [Reactant & Reagent Setup]: State reactants and specific reagents (e.g. $CHCl_3$ + NaOH) .

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Explain the preparation of alcohols by hydration of alkenes (acid-catalyzed hydration mechanism). [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Step 1 (Protonation)]** Alkene attacks H_3O^+ electrophile: $CH_2=CH_2 + H^+ <= => CH_3-CH_2^+$ (carbocation intermediate). [1 Mark]

• **[Step 2: Step 2 & 3 (Nucleophilic attack & Deprotonation)]** H_2O attacks carbocation: $CH_3-CH_2^+ + H_2O: <= => CH_3-CH_2-OH_2^+$ (oxonium ion). Loss of proton: $CH_3-CH_2-OH_2^+ + H_2O <= => CH_3-CH_2-OH + H_3O^+$. Follows Markovnikov's rule for unsymmetrical alkenes. [2 Marks]

• **Key Exam Result:** Acid-catalyzed hydration via carbocation intermediate (Markovnikov addition)

2. Explain the preparation of alcohols from Grignard reagents by reaction with methanal, other aldehydes, and ketones. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Reactions]** 1. HCHO + RMgX --> $R-CH_2-OMgX$ --> $[H_3O^+]$ $R-CH_2-OH$ (Primary 1° alcohol). 2. R'CHO + RMgX --> R'-CH(R)-OMgX --> $[H_3O^+]$ R'-CH(OH)-R (Secondary 2° alcohol). 3. $R'_2C=O$ + RMgX --> $R'_2C(R)-OMgX$ --> $[H_3O^+]$ $R'_2C(OH)R$ (Tertiary 3° alcohol). [3 Marks]

• **Key Exam Result:** HCHO --> 1° alcohol | RCHO --> 2° alcohol | Ketones --> 3° alcohol

3. Describe the commercial preparation of phenol from cumene (cumene hydroperoxide process) with equations. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Oxidation of Cumene]** $C_6H_5-CH(CH_3)_2 + O_2$ (air) --> $C_6H_5-C(CH_3)_2-O-O-H$ (cumene hydroperoxide). [1.5 Marks]

• **[Step 2: Acid Cleavage]** Cumene hydroperoxide + dilute H_2SO_4 --> C_6H_5-OH (phenol) + $CH_3-CO-CH_3$ (acetone). Economically dominant route because acetone is obtained as a valuable coproduct in high yield. [1.5 Marks]

• **Key Exam Result:** Cumene --> Cumene hydroperoxide --> Phenol + Acetone

4. Write chemical equations and mechanisms for acid-catalyzed dehydration of ethanol to ethene at 443 K. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Temperature Dependency]** At 443 K (conc. H_2SO_4): CH_3CH_2OH --> $CH_2=CH_2$ (ethene) + H_2O (Intramolecular elimination). At 413 K (excess alcohol): 2 CH_3CH_2OH --> $C_2H_5-O-C_2H_5$ (diethyl ether) + H_2O (Intermolecular substitution). [2 Marks]

• **[Step 2: Mechanism at 443 K]** 1. Protonation: $CH_3CH_2OH + H^+ <= => CH_3CH_2-OH_2^+$. 2. Loss of H_2O: $CH_3CH_2-OH_2^+$ --> $CH_3CH_2^+$ (slow). 3. Elimination of β-H: $CH_3CH_2^+$ --> $CH_2=CH_2 + H^+$. [2 Marks]

• **Key Exam Result:** 443 K gives Ethene (elimination) | 413 K gives Diethyl Ether (substitution)

5. Explain Kolbe's reaction and Reimer-Tiemann reaction of phenol with balanced chemical equations. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Kolbe's Reaction]** Phenol + NaOH --> Sodium phenoxide + CO_2 (400 K, 4-7 atm) --> sodium salicylate --> $[H^+]$ Salicylic acid (2-hydroxybenzoic acid). [2 Marks]

• **[Step 2: Reimer-Tiemann Reaction]** Phenol + $CHCl_3$ + 3 NaOH (aq, 340 K) --> intermediate with $-CHCl_2$ at ortho --> [NaOH] --> $[H^+]$ Salicylaldehyde (2-hydroxybenzaldehyde). Electrophile is dichlorocarbene ($:CCl_2$). [2 Marks]

• **Key Exam Result:** Kolbe: Salicylic acid | Reimer-Tiemann: Salicylaldehyde (electrophile $:CCl_2$)

6. Explain why the boiling points of alcohols are much higher than ethers and hydrocarbons of comparable molecular mass. **[Score: 2 Marks]**

Answer & Key Valuation Steps:

• **[Step 1: Hydrogen Bonding]** Alcohols contain highly electronegative oxygen bonded to hydrogen (-O-H), allowing strong intermolecular hydrogen bonding that holds molecules in vast associated networks. Ethers and hydrocarbons lack O-H bonds and can only form weak dipole-dipole or London dispersion forces. Hence alcohols boil at vastly higher temperatures. [2 Marks]

• **Key Exam Result:** Extensive intermolecular H-bonding elevates alcohol boiling points

7. Write the structure of the products formed when anisole (methoxybenzene) is treated with: (a) Br2 in ethanoic acid, (b) conc. HNO3/H2SO4. **[Score: 2 Marks]**

Answer & Key Valuation Steps:

• **[Step 1: Products]** Methoxy group (-OCH_3) is strongly ortho/para directing. (a) Bromination with Br_2 divided by CH_3COOH: p-bromoanisole (major, 90%) + o-bromoanisole (minor). (b) Friedel-Crafts alkylation with CH_3Cl divided by anhydrous $AlCl_3$: 4-methoxytoluene (major) + 2-methoxytoluene. [2 Marks]

• **Key Exam Result:** p-bromoanisole (major) and 4-methoxytoluene (major)